

A **Non-Parametric Hypothesis Test**, also known as a distribution-free method, is a statistical procedure that assesses whether a hypothesis is true or false without making assumptions about the distribution of the data. Non-parametric tests are used when the assumptions of a parametric test are not met. Can adapt to ordinal and categorical data.

Some non-parametric tests are: Chi-square test, Sign test, Run test, Kruskal-Wallis test, Mann-Whitney-Wilcoxon test, and Kolmogorov-Smirnov test.

A chi-square test is a statistical hypothesis test that determines if data is significantly different from what is expected. It compares **observed values** in data to the **expected values if the null hypothesis is true**.

- SPSS Tutorials: Chi-Square Test of Independence - LibGuides

10 Jul 2024 — This test utilizes a contingency table to analyze the data. A contingency table (also known as a cross-tabulation, cros...

Kent State University

Example:

A University conducted a survey on health awareness among its students. The survey revealed that only a minor proportion of 15% of students do physical exercise/ yoga regularly, whereas 25% do physical exercise/ yoga sporadically, and rest 60% do not do any exercise/ yoga.

The University launched a health promotion campaign on campus. To evaluate the impact of the program, the University again conducted a survey.

Out of 470 students surveyed, ed graduates and asked the same questions. The survey was completed by 470 graduates and the following data as obtained.

	No Exercise	Sporadic Exercise	Regular Exercise	Total
Observation	255	125	90	470

Has the health promotion campaign caused any change in behaviour of students towards exercise/yoga?

Hypotheses and level of significance.

Null Hypothesis H_0 – There is no change.

Alt. Hypothesis H_a -- Null hypothesis is false.

Predict the outcome of survey at 5% level of significance.

	No Exercise	Sporadic Exercise	Regular Exercise	Total
Observation	255	125	90	470
Expected	60%	25%	15%	
	282.0	117.5	70.5	

The test statistic is:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$\chi^2 = \frac{(255-282)^2}{282} + \frac{(125-117.5)^2}{117.5} + \frac{(90-70.5)^2}{70.5}$$

$$\chi^2 = 2.59 + 0.48 + 5.39 = 8.46$$

Decision Rule to accept or reject Null hypothesis will depend on level of significance and the degrees of freedom.

Critical values can be found in a table of probabilities for the χ^2 distribution. Here we have $df=k-1=3-1=2$ and a 5% level of significance. The appropriate critical value is 5.99, and the decision rule is as follows: Reject H_0 if $\chi^2 \geq 5.99$.

Example 2:

Data on the distribution of weight (in categories) based on BMI index among people in 2002 were as follows: 2% Underweight, 39% Normal Weight, 36% Overweight, and 23% Obese. Suppose we want to assess whether over the years the distribution of BMI has changed. Using data from the $n=3,326$ participants observed the following:

Under weight BMI <18.5	Normal Weight BMI 18.5-24.9	Over weight BMI 25.0-29.9	Obese BMI ≥ 30	Total
20	932	1374	1000	3326

H_0 : Distribution of responses is 0.02, 0.39, 0.36, 0.23

H_1 : H_0 is false. $\alpha=0.05$

	Under weight BMI <18.5	Normal Weight BMI 18.5-24.9	Over weight BMI 25.0-29.9	Obese BMI ≥ 30	Total
O	20	932	1374	1000	3326
E	66.5	1297.1	1107.4	765	

$$\chi^2 = \frac{(20-66.5)^2}{66.5} + \frac{(932-1297.4)^2}{1297.4} + \frac{(1374-1197.4)^2}{1197.4} + \frac{(1000-765.0)^2}{765.0}$$

$$\chi^2 = 32.52 + 102.77 + 26.05 + 72.19 = 233.53$$

Decision rule.

df=k-1=4-1=3 and 5% level of significance ($\alpha=0.05$). The appropriate critical value is 7.81

Since $\chi^2 > 7.81$, Reject H_0

Advantages of the Chi-Square Test:

Flexibility	Non-parametric Nature:
Ease of Calculation	Versatility
Categorical data	Large Sample Sizes:

Limitations of the Chi-Square Test:

Sample Size Sensitivity Requirement	Expected Frequency
Independence Assumption	Non-parametric Nature
Categorical Data Limitation	

Doesn't provide information about strength of the relationship or its nature

Two types of Chi-Square Tests

1. Chi-Square Goodness-of-Fit Test

- To determine if a sample data matches a population with a specific distribution.

2. Chi-Square Test of Independence

- To determine if there is a significant association between categorical variables. e.g. Checking if gender and preference for a new product are related.

Example:

	No Regular Exercise	Sporadic Exercise	Regular Exercise	Total
Out Campus Hostel	32	30	28	90
In Campus Hostel	74	64	42	180
Home Apartments	110	25	15	150
Home Single House	39	6	5	50
Total	255	125	90	470

Hypotheses and level of significance.

H_0 : Living arrangement has no effect on exercise

H_1 : H_0 is false. $\alpha=0.05$

	No Regular Exercise	Sporadic Exercise	Regular Exercise	Total
Out Campus Hostel	32 (48.8)	30 (23.9)	28 (17.2)	90
In Campus Hostel	74 (97.7)	64 (47.9)	42 (34.5)	180
Home Apartments	110	25	15	150

	(81.4)	(39.9)	(28.7)	
Home Single House	39 (27.1)	6 (13.3)	5 (9.6)	50
Total	255	125	90	470

$$\chi^2 = \frac{(32-48.8)^2}{48.8} + \frac{(30-23.9)^2}{23.9} + \frac{(28-17.2)^2}{17.2} \\ + \frac{(74-48.8)^2}{48.8} + \frac{(64-47.9)^2}{47.9} + \frac{(42-34.5)^2}{34.5} \\ + \frac{(110-81.4)^2}{81.4} + \frac{(25-39.9)^2}{39.9} + \frac{(15-28.7)^2}{28.7} \\ + \frac{(39-27.1)^2}{27.1} + \frac{(6-13.3)^2}{13.3} + \frac{(5-9.6)^2}{9.6}$$

$$\chi^2 = 5.78 + 1.58 + 6.78 + 5.75 + 5.41 + 1.63 \\ + 10.05 + 5.58 + 6.54 + 5.23 + 4.01 + 2.20 = 60.5$$

For two way data table

df = (r-1)(c-1), where r and c are numbers of rows and columns.

df = (4-1)(3-1) = 6.

For df=6 and 5% level of significance ($\alpha = 0.05$),

Critical value χ^2 is 12.59.

Since $60.5 \geq 12.59$, Reject H_0

Prob:

	No Regular Exercise	Sporadic Exercise	Regular Exercise	Total
Out Campus Hostel	32	30	28	90
In Campus Hostel	74	64	42	180

Is there any difference between Out Campus Hostel and In Campus Hostel?

Sign Test

It is a simple nonparametric test for matched or paired data.

Child	Before Treatment	After 1 Week of Treatment	Difference (Before-After)	Sign
1	85	75	10	+
2	70	50	20	+
3	40	50	-10	-
4	65	40	25	+
5	80	20	60	+
6	75	65	10	+
7	55	40	15	+
8	20	25	-5	-

H_0 : There is no difference due to treatment

H_1 : H_0 is false. $\alpha=0.05$

The approach is to analyze only the signs of the difference scores

Test statistic is number of positive signs or number of negative signs, whichever is smaller. $C = \min(2, 6) = 2$

Critical value

For Sample size, number of matched pairs i.e. $n=8$
and level of significance $\alpha=0.05$.

Critical value is 1 (From Sign Test Table)

Since $2 > 1$, Do not reject H_0

The Critical Value can be calculated using Binomial Theorem

Example:

S.No.	Control	Treatment	Sign
-------	---------	-----------	------

1	7.95	10.3	+
2	7.05	7.97	+
3	4.73	4.83	+
4	4.96	5.51	+

H_0 : There is no improvement

Test with 5% level of significance

If null hypothesis is true, probability of improvement $p=0.5$ and no improvement $q = 1- p = 0.5$

In 4 trials (samples), all are success.

Prob of 4 successes in 4 trials

$$P(x \text{ successes}) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$

Here $n = 4$, $x = 4$, and $p = 0.5$

So $P = 1 \times (0.5)^4 \times (0.5)^0 = 0.0625$

Since $P > 0.05$, The Null hypothesis is not rejected. We need more number of trials /samples.

If there is no effect

Prob. of no Improvement $p = 0.5$ and improvement $q = (1- 0.5) = 0.5$

$$P(x \text{ successes}) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$

Probability of success in 8 trials

$n = 8$, $p = 0.5$, $x = 0, 1, 2, \dots 7, 8$

0	0.0039	3	0.2188	6	0.1094
1	0.0313	4	0.2734	7	0.0313
2	0.1094	5	0.2188	8	0.0039

Number of Improvement (n)	Prob. P(n)	Cum. Prob. P(x≤n)	Prob. of Improvement	Prob. of no Improvement P(x>n)
0	0.004	0.004	0.004	0.996
1	0.031	0.035	0.035	0.965

2	0.109	0.145	0.145	0.855
3	0.219	0.363	0.363	0.637
4	0.273	0.637	0.637	0.363
5	0.219	0.856	0.856	0.144
6	0.109	0.965	0.965	<u>0.034</u>
7	0.031	0.996	0.996	0.004
8	0.004	1	1	0.0

Number of Sign Change	Prob.	Prob. of Improvement	Prob. of no difference
0	0.008	$P(x=0)$ or $P(x=8)$	0.008
1	0.062	$P(x \leq 1)$ or $P(x \geq 7)$	0.07
2	0.218	$P(x \leq 2)$ or $P(x \geq 6)$	0.288
3	0.438	$P(x \leq 3)$ or $P(x \geq 5)$	0.726
4	0.273	$P(x \leq 4)$ or $P(x \geq 4)$	1.0

$$P(x \leq 1) = P(0) + P(1) = 0.008 + 0.031 = 0.035$$

Limitations of the Sign Test:

1. **Loss of Information:** Considers the direction of changes (positive or negative). This can result in a loss of valuable information, if the magnitude is important to the analysis.
2. **Low Power:** Less powerful compared to parametric tests (like the paired t-test). Requires a larger sample size for same level of significance.
3. **Ties Handling:** In cases where there are a lot of ties (no change between pairs), the sign test may not be effective.
4. **Simple Hypotheses Only:** It does not allow for more complex hypotheses or comparisons involving variances or distributions.

Applicability: Best used when data does not meet the assumptions for parametric tests, such as normality or equal variances. For datasets

that meet these assumptions, parametric tests are generally preferred for their greater power and sensitivity.

One sided vs two sided test

If '+' indicates improvement H_0 : No improvement is rejected at $\alpha = 0.05$ If $x \geq N$ H_0 is rejected	H_0 : No Difference is rejected at $\alpha = 0.05$ If $x \geq N_1$ and $x \leq N_2$ the H_0 is rejected
If $x \geq 7$ H_0 : No improvement is rejected at $\alpha = 0.05$	If $(x \leq 1)$ or $P(x \geq 7)$ $0.035 + 0.035 = 0.07$ H_0 : No difference <u>cannot be rejected</u> at $\alpha = 0.05$

RUN Test

The runs test is a statistical test to determine whether the data is random or whether there is a pattern.

The runs test is a type of non-parametric test and hence, there is no need for the assumption of a normal distribution to hold true.

Example:

F F M M F M F F F F M F F M M M M M F F

Runs : $\underbrace{F F}_1 \underbrace{M M}_2 \underbrace{F}_3 \underbrace{M}_4 \underbrace{F F F F}_5 \underbrace{M}_6 \underbrace{F F}_7$

Runs : $\underbrace{M M M M M}_8 \underbrace{F F}_9$

Total $n = 20$ Female (n_1) = 11, Male (n_2) = 9 runs (r) = 9.

The normal distribution via the following formula:

$$z = (r - \mu_r) / \sigma_r$$

where:

- r is the actual number of runs;
- μ_r is the expected number of runs; and
- σ_r is the standard deviation of the number of runs.

The values of μ_r and σ_r are computed as follows:

$$\mu_r = \frac{2n_1n_2}{n_1 + n_2} + 1$$

$$\sigma_r = \sqrt{\frac{(2n_1n_2)(2n_1n_2 - n_1 - n_2)}{(n_1 + n_2)^2 (n_1 + n_2 - 1)}}$$

Let's calculate for mean and standard deviation of the runs:

$$\mu_r = \frac{2n_1n_2}{n_1+n_2} + 1 \quad \mu_r = \frac{2 * 11 * 9}{11 + 9} + 1 = 10.9$$

And

$$\sigma_r = \sqrt{\left[\frac{(2 * 11 * 9)(2 * 11 * 9 - 11 - 9)}{(11 + 9)^2 (11 + 9 - 1)} \right]} = 2.15$$

$$z = (r - \mu_r) / \sigma_r = (9 - 10.9) / 2.15 = -0.884$$

At the 5% significance level, a test statistic with an absolute value greater than 1.96 indicates non-randomness (reject H_0).

Example: Blood test samples. The data-set are as follow:

7.7, 10.0, 8.8, 12.1, 7.4, 6.1, 6.1, 6.1, 8.0, 8.8,
6.1, 10.9, 5.0, 12.6, 11.7, 11.5, 6.4, 8.9, 7.2, 7.2,
6.5, 10.3, 7.9, 7.8, 14.0, 6.8, 7.7, 7.8, 6.7, 7.7

Numerical data is converted to categorical data by grouping using a cut-off point. The cut-off points can be mean, median, mode or by custom. By custom, let us take 7 as cutoff point to categorise data into two groups, good (G) or poor (P).

The data are grouped as follows:

PPPPP GGG PP G P G PPP G PPP G
PPPP G PP G P

$$n_1 = 21 \quad n_2 = 9$$

Using equations for μ_r and σ_r

We get $\mu_r = 13.6$ and $\sigma_r = 2.245$

Number of runs $r = 15$

So $z = (r - \mu_r) / \sigma_r = (15 - 13.6) / 2.245 = 0.624$

At the 5% significance level, a test statistic with an absolute value greater than 1.96 indicates non-randomness.

Since test statistics < 1.96 , the data is random, i.e. No pattern is detected

Kruskal-Wallis Test

The Kruskal Wallis test is used to compare **medians among k comparison groups** ($k > 2$) and is sometimes described as an ANOVA with the data replaced by their ranks.

The procedure for the test involves pooling the observations from the k samples into one combined sample in order of rank.

Example: A clinical study is designed to assess differences in albumin levels in adults following diets with different amounts of protein. Three diets are compared, ranging from 5% to 15% protein. The albumin levels of participants following each diet are shown below.

5% Protein	10% Protein	15% Protein
3.1	3.8	4.0
2.6	4.1	5.5
2.9	2.9	5.0
	3.4	4.8
	4.2	

Is there is a **significant** difference in serum albumin levels among subjects on the three different diets.

Sample sizes are small and not equal across comparison groups ($n_1=3$, $n_2=5$, $n_3=4$).

H_0 : There is no difference between three population medians

To conduct the test, we first order the data in the combined total sample of 12 subjects from smallest to largest. We also need to keep track of the group assignments in the total sample.

			Total Sample (Ordered Smallest to Largest)	Ranks
--	--	--	--	-------

T1: 5% Protein	T2: 10% Protein	T3: 15% Protein	5% Protein	10% Protein	15% Protein	5% Protein	10% Protein	15% Protein
3.1	3.8	4.0	2.6			1		
2.6	4.1	5.5	2.9	2.9		2.5	2.5	
2.9	2.9	5.0	3.1			4		
	3.4	4.8		3.4			5	
	4.2			3.8			6	
					4.0			7
				4.1			8	
				4.2			9	
					4.8			10
					5.0			11
					5.5			12
						7.5	30.5	40

Test Statistic for the Kruskal Wallis Test

The test statistic for the Kruskal Wallis test is denoted H and is defined as follows:

$$H = \left(\frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} \right) - 3(N+1)$$

where k=the number of comparison groups, N= the total sample size, n_j is the sample size in the j^{th} group and R_j is the sum of the ranks in the j^{th} group.

In this example $R_1 = 7.5$, $R_2 = 30.5$, and $R_3 = 40$.

The H statistic for this example is computed as follows: .

$$H = \left(\frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} \right) - 3(N+1) = \frac{12}{12(13)} \left(\frac{7.5^2}{3} + \frac{30.5^2}{5} + \frac{40^2}{4} \right) - 3(13) = 7.52$$

Compare with critical value of H.

If $H \geq$ critical value, reject H_0 in favour of H_1

From **Kruskal Wallis Test table** the critical value is 5.656.

Since $7.52 \geq 5.656$, Reject H_0

Example:

A personal trainer is interested in comparing the anaerobic thresholds of athletes. Anaerobic threshold is important parameter for upper limit of aerobic exercise. The following data are anaerobic thresholds for distance runners, distance cyclists, distance swimmers and cross-country skiers.

Distance Runners (A)	Distance Cyclists (B)	Distance Swimmers (C)	Cross-Country Skiers (D)
185	190	166	201
179	209	159	195
192	182	170	180
165	178	183	187
174	181	160	215

Is there a difference in anaerobic thresholds among the different groups of athletes?

Step 1. Set up hypotheses and determine level of significance.

H_0 : There is no difference in median score of four population

Test with level of significance $\alpha=0.05$

Step 2. Select the appropriate test statistic.

The test statistic for the Kruskal Wallis test is denoted H and is defined as:

$$H = \left(\frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} \right) - 3(N+1)$$

Step 3. Compute the test statistic.

Distance Runners (A)		Distance Cyclists (B)		Distance Swimmers (C)		Cross-Country Skiers (D)	
165	3	178	7	159	1	180	9
174	6	181	10	160	2	187	14
179	8	182	11	166	4	195	17
185	13	190	15	170	5	201	18

192	16	209	19	183	12	215	20
	R ₁ =46		R ₂ =62		R ₃ =24		R ₄ =78

$$H = \left(\frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} \right) - 3(N+1) = \frac{12}{20(21)} \left(\frac{46^2}{5} + \frac{62^2}{5} + \frac{24^2}{5} + \frac{78^2}{5} \right) - 3(21) = 9.11$$

Step 4. Compare with Critical Values of H

For the Kruskal Wallis, if sample size is 5 or more in each group, the test statistic H approximates a chi-square distribution with df=k-1.

From Chi Square table critical value for df = 3 and $\alpha=0.5$ is 7.81

Since $9.11 \geq 7.81$, Reject H_0